Numerical Pricing of European Calls via Method of Lines
MATH/CS 514 Final Project

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Abstract

This report prices European call options under the Black–Scholes (BS) model with continuous dividend yield using a Method of Lines (MOL) discretization. The BS PDE is semi-discretized in the asset-price dimension with central differences, producing a stiff ODE system integrated by implicit Backward Euler and BDF2 schemes (Chapter 12 of Süli & Mayers). Convergence is assessed against the closed-form BS solution, and a small market-data case study illustrates the effect of dividend corrections. All code lives in `BS_Solver.py` and the accompanying Jupyter notebooks; the repository is on Github.

1 Physical Phenomenon and Model

A European call option is a contract that gives its holder the right (but not the obligation) to buy an underlying asset at a fixed strike K on a maturity date T. Let S_t denote the asset price at time t. Under the usual Black–Scholes assumptions [2]—frictionless trading, continuous rebalancing, constant risk-free rate r, constant volatility σ, and a continuous dividend yield q—the discounted option price can be written as a risk-neutral expectation

$$V(S_0, 0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[\max(S_T - K, 0)\big],$$

where $\mathbb{Q}$ is the risk-neutral measure under which S_t follows a geometric Brownian motion with drift $r - q$ and volatility σ.

By standard arbitrage arguments or via the Feynman–Kac formula, the option price $V(S, t)$ solves the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q)S\frac{\partial V}{\partial S} - rV = 0,$$

for $0 < t < T$ and $S > 0$, with terminal payoff

$$V(S, T) = \max(S - K, 0).$$

For numerical convenience I reverse time by introducing the time-to-maturity variable $\tau = T - t$. In terms of (S, τ) the PDE becomes

$$\frac{\partial V}{\partial \tau} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q)S\frac{\partial V}{\partial S} - rV,$$

*Department of Mathematics, University of Wisconsin-Madison. yyang773@wisc.edu. I thank Professor Laurel Ohm for her helpful comments for this manuscript. All errors are my own. Click here to see Repository: github.com/franxyang/MATH514FinalProject-BSModel.

posed for $0 < \tau \leq T$ with initial condition at $\tau = 0$,

$$V(S, 0) = \max(S - K, 0).$$

On a truncated domain $[0, S_{\max}]$ I impose an absorbing boundary at the origin and a linear growth boundary at the far field,

$$V(0, \tau) = 0, \qquad V(S_{\max}, \tau) = S_{\max} e^{-q\tau} - K e^{-r\tau},$$

which reflects the large-S asymptotics of a call option.

A logarithmic change of variables $x = \ln S$ and the time reversal above transform the PDE into a constant-coefficient diffusion (a heat equation) on x, which explains the smoothing behavior of the solution and motivates the use of implicit time-stepping schemes for stability. The solution V and its sensitivities with respect to S (Delta and Gamma) are the quantities traders use in practice to construct hedging strategies; later sections show how these are approximated numerically by the Method of Lines.

2 Numerical Method (MOL)

After spatial discretization, the Black–Scholes PDE on $[0, S_{\max}]$ is reduced to a finite-dimensional ODE system

$$V'(\tau) = AV(\tau) + b(\tau),$$

where $V(\tau) \in \mathbb{R}^{M-1}$ collects the interior nodal values $V(S_i, \tau)$ for $i = 1, \ldots, M-1$, $A \in \mathbb{R}^{(M-1)\times(M-1)}$ is a tridiagonal matrix, and $b(\tau)$ encodes the contribution of the right boundary. This yields a stiff initial-value problem of the type treated in our textbook.

Spatial semi-discretization.

I use a uniform grid $S_i = i\,\Delta S$ on $[0, S_{\max}]$ with $\Delta S = S_{\max}/M$. Central differences approximate the first and second derivatives,

$$\frac{\partial V}{\partial S}(S_i, \tau) \approx \frac{V_{i+1}(\tau) - V_{i-1}(\tau)}{2\,\Delta S}, \qquad \frac{\partial^2 V}{\partial S^2}(S_i, \tau) \approx \frac{V_{i+1}(\tau) - 2V_i(\tau) + V_{i-1}(\tau)}{\Delta S^2},$$

which gives a second-order accurate spatial discretization in ΔS. The right boundary condition $V(S_{\max}, \tau) = S_{\max} e^{-q\tau} - K e^{-r\tau}$ enters via the top row of the tridiagonal stencil and contributes to $b(\tau)$. Increasing M improves spatial resolution and Greeks but also enlarges the system and increases stiffness; choosing $S_{\max}$ too small truncates deep in-the-money prices, while choosing it too large can worsen conditioning without much benefit.

Time-stepping.

The semi-discrete system $V' = AV + b(\tau)$ is stiff because the largest eigenvalues of A scale like $\mathcal{O}((\Delta S)^{-2})$. Explicit schemes would require prohibitively small time steps for stability. Instead, I use an implicit Backward Euler (BE) step to start, followed by the two-step BDF2 method for higher accuracy. For a time grid $\tau_n = n\,\Delta\tau$ with $n = 0, \ldots, N$, BE and BDF2 take the form

$$(I - \Delta\tau\, A)V^{n+1} = V^n + \Delta\tau\, b^{n+1} \quad \text{(Backward Euler)},$$

$$V^{n+1} - \tfrac{4}{3}V^n + \tfrac{1}{3}V^{n-1} = \tfrac{2}{3}\Delta\tau\left(AV^{n+1} + b^{n+1}\right) \quad \text{(BDF2)}.$$

At each step I solve the resulting sparse linear system using SciPy's sparse solvers. Backward Euler is L-stable and robust for the first step; BDF2 is A-stable and second-order accurate in $\Delta\tau$, making the BE–BDF2 combination well suited to this stiff MOL system. The analytic Black–Scholes formula is used as a reference solution; to avoid singularities at expiry, the reference function treats $\tau = 0$ explicitly via the payoff and only evaluates the closed-form formula for $\tau > 0$.

3 Convergence Study

Closed-form Black–Scholes prices provide a convenient reference solution for assessing the accuracy of the MOL scheme. The analytic function treats $\tau = 0$ explicitly via the payoff $V(S, 0) = \max(S - K, 0)$ and only evaluates the Black–Scholes formula for $\tau > 0$, which avoids singularities at expiry while keeping the reference solution smooth in time.

I first study temporal convergence for BDF2 on a pinned synthetic case with

$$S_{\max} = 400, \quad K = 100, \quad T = 1, \quad r = 0.03, \quad \sigma = 0.20, \quad q = 0.01, \quad M = 400.$$

Here M is fixed so that spatial error is non-negligible, and I refine only the number of time steps N. Let $V_h(S_i, T)$ denote the numerical solution at maturity and $V_{\text{ref}}(S_i, T)$ the reference Black–Scholes value on the same grid. The discrete L^2 error is defined by

$$\|e\|_2^2 = \Delta S \sum_{i=1}^{M-1} \big(V_h(S_i, T) - V_{\text{ref}}(S_i, T)\big)^2.$$

For successive refinements $N \mapsto 2N$ with time steps $\Delta\tau_N = T/N$, the observed temporal order is

$$p_N \approx \log_2 \frac{\|e_N\|_2}{\|e_{2N}\|_2},$$

where $\|e_N\|_2$ is the error measured with N steps. The results are summarized in Table 1, and Figure 1 shows the corresponding log–log plot of error versus $\Delta\tau$.

N	$\Delta\tau$	L^2 error	Observed order
50	0.0200	7.108×10^{-4}	–
100	0.0100	6.482×10^{-4}	0.13
200	0.0050	6.329×10^{-4}	0.03
400	0.0025	6.291×10^{-4}	0.01

Table 1: Temporal errors vs. time steps (BDF2, $M = 400$). Spatial discretization error dominates once $\Delta\tau$ is sufficiently small, flattening the observed order.

The error plateau in Table 1 and Figure 1 shows that spatial discretization error dominates for these parameter choices: refining $\Delta\tau$ beyond a certain point no longer reduces the total error, so the observed order drops toward zero even though BDF2 is second-order accurate in time. To see the interplay between spatial and temporal errors more clearly, I perform a complementary spatial study (Figure 2) in which N is fixed and M is varied. As M increases, the saturated error level decreases, illustrating that recovering the theoretical second-order temporal convergence requires balancing M, N, and $S_{\max}$ so that both spatial and temporal errors are in the asymptotic regime.

4 Greeks and Solution Behavior

Greeks quantify sensitivities of the option value to underlying parameters and are central in hedging. In this project I focus on Delta and Gamma with respect to the underlying price S.

Finite-difference Greeks

Given the numerical solution $V(S_i, T)$ on the spatial grid $\{S_i\}_{i=0}^{M}$, I approximate Delta and Gamma at maturity by centered finite differences on the same grid:

$$\Delta(S_i) \approx \frac{V(S_{i+1}, T) - V(S_{i-1}, T)}{2\,\Delta S}, \qquad \Gamma(S_i) \approx \frac{V(S_{i+1}, T) - 2V(S_i, T) + V(S_{i-1}, T)}{\Delta S^2},$$

for interior indices $i = 1, \ldots, M-1$. These stencils are second-order accurate in ΔS and naturally reuse the discrete structure of the MOL scheme. At the boundaries I either use one-sided differences or simply omit the plotted values. For validation, the numerical Greeks can be compared against the analytic Black–Scholes Greeks computed from the closed-form solution at the same grid points.

Qualitative behavior

For a representative parameter set with $M = 1000$ and $N = 500$, the numerical Delta transitions smoothly from 0 in the deep out-of-the-money region to 1 in the deep in-the-money region, with a steep slope near $S \approx K$. Gamma exhibits a single pronounced peak around the strike, reflecting the region where the option value is most curved with respect to S. As the grid is refined (increasing M), this Gamma peak becomes sharper and closer to the analytic profile; on coarser meshes, the peak is smeared out and its height is underestimated, which corresponds to underestimating the convexity of the option value.

Figure 3 displays the numerical price, Delta, and Gamma for $M = 1000$, $N = 500$. The plots illustrate that the MOL discretization together with the implicit BE–BDF2 time-stepping is able to reproduce the expected qualitative behavior of both the option price and its sensitivities, provided the spatial grid is sufficiently fine near the strike.

5 Market Case Study (Offline Snapshot)

To illustrate the behavior of the MOL solver on real-looking data, I run a small offline case study based on a cached option chain snapshot (approximate Jan 2026 expiries). Live data access is disabled in the project environment, so the underlying spot prices, strikes, implied volatilities, and dividend yields are stored in a local JSON file and treated as fixed inputs.

Data and parameter choice.

For each ticker, the spot price S_0 and strike K come from a single near-the-money call in the chain. The time to maturity $T \approx 0.60$ years is computed from the quoted expiry date. The risk-free rate $r = 4.2\%$ is a simple proxy for the prevailing short-term USD yield at the snapshot date. The volatility σ is taken as the mid implied volatility of that option (averaging bid and ask IVs), and the continuous dividend yield q is an approximation based on recent trailing dividends converted to a continuously compounded rate. These inputs are then fed into the same MOL solver used in the synthetic tests.

I price each contract with the BDF2 scheme on a relatively fine grid ($M = 1000$, $N = 200$) both without dividends ($q = 0$) and with the calibrated dividend yield $q > 0$. The resulting PDE prices are compared to the observed market mids in Table 2.

Table 2: Offline snapshot-BS PDE prices vs. market mid ($T \approx 0.60$ yr, $r = 4.2\%$, $M = 1000$, $N = 200$).

Ticker	S_0	K	σ	q	PDE no-div	PDE with-div
AAPL	170.12	170.0	0.28	0.0050	16.78	16.48
MSFT	370.44	370.0	0.26	0.0080	34.40	33.36
GOOGL	136.21	135.0	0.24	0.0000	12.37	12.37
NVDA	480.55	480.0	0.45	0.0005	72.06	71.97

The corresponding market mids for these contracts were {8.55, 14.10, 5.60, 36.80}, so the simple European Black–Scholes PDE model materially overprices the options in this snapshot. This is not surprising: the model ignores early-exercise features (American or employee-style contracts), volatility smiles and term structures, discrete dividend timings, and any risk premia or microstructure effects. Moreover, the calibration here is deliberately minimal (a single σ and q per contract, without fitting to an implied-volatility surface).

Nevertheless, the comparison is informative. Including a positive dividend yield q consistently lowers the model price for dividend-paying names (AAPL, MSFT), nudging the PDE prices toward the observed mids, while having negligible effect for near-non-dividend payers (GOOGL, NVDA). This small case study illustrates how the same MOL solver that was validated on synthetic data can be used as a building block in more realistic option-pricing workflows, provided that richer models (American exercise, calibrated volatility surfaces, more accurate dividend modeling) are added on top.

6 Discussion and Limitations

The experiments highlight several numerical and modeling trade-offs.

- **Stability vs. accuracy.** The MOL discretization leads to a stiff ODE system $V' = AV + b(\tau)$ whose eigenvalues scale like $\mathcal{O}((\Delta S)^{-2})$. Implicit Backward Euler and BDF2 are A-stable and therefore well suited to this regime, but the temporal convergence study shows that once $\Delta\tau$ is sufficiently small, the total error is dominated by the spatial discretization (Table 1 and Figure 1). In practice, balancing M, N, and $S_{\max}$ so that both spatial and temporal errors are in their asymptotic regimes is essential.
- **Boundary and grid choice.** The upper boundary $S_{\max}$ must be chosen large enough that the linear asymptotic condition $V(S_{\max}, \tau) \approx S_{\max} e^{-q\tau} - Ke^{-r\tau}$ is valid for the options of interest. Too small an $S_{\max}$ truncates deep in-the-money values and biases prices and Greeks; too large an $S_{\max}$ can worsen conditioning and increase cost without much benefit. The spatial study (Figure 2) suggests that adaptive choices of $S_{\max}$ and possibly nonuniform grids concentrated near the strike could reduce error for a given number of degrees of freedom.
- **Greeks and mesh resolution.** Finite-difference Greeks inherit the same spatial resolution constraints as the underlying solution. Gamma in particular is sensitive to ΔS; coarse grids smear the peak near $S \approx K$ and underestimate curvature, while finer grids (e.g. $M = 1000$) recover the expected shape (Figure 3). For applications where Greeks drive hedging decisions, adaptive or nonuniform meshes would be especially valuable.
- **Model fidelity.** The market snapshot in Table 2 shows that even with a reasonably fine

grid, the calibrated European Black–Scholes PDE overprices the observed calls. This reflects structural model limitations rather than numerical error: real contracts may be American, implied volatilities vary across strikes and maturities (smiles and surfaces), and dividends are paid discretely rather than as a smooth yield. Extending the framework to early exercise (e.g. via projected SOR or penalty methods), fitting a volatility surface, or moving to stochastic-volatility models such as Heston would address some of this bias.

7 Conclusion

The Method of Lines provides a clean bridge between the Black–Scholes pricing PDE and the stiff ODE systems studied in our textbook. By discretizing in the asset-price variable and applying implicit Backward Euler for startup and BDF2 thereafter, the resulting solver is stable for realistic parameter ranges and can be implemented efficiently using sparse linear algebra.

Synthetic tests against the analytic Black–Scholes formula confirm the expected behavior: temporal errors for BDF2 decrease with $\Delta\tau$ until spatial discretization error dominates, and finer spatial grids improve both prices and Greeks. The finite-difference Delta and Gamma reproduce the standard qualitative profiles provided the mesh is sufficiently refined near the strike.

An offline market case study (Table 2) shows that incorporating continuous dividend yields systematically lowers prices for dividend payers and nudges them toward observed mids, but also highlights the limits of the plain European Black–Scholes model in capturing real option prices. Overall, the project demonstrates how the MOL + implicit ODE framework can be used to price options and compute sensitivities in a numerically robust way, while pointing toward natural extensions: calibrated volatility surfaces, early-exercise features, more realistic dividend modeling, and adaptive or nonuniform spatial grids to sharpen Gamma near the strike.

Figures

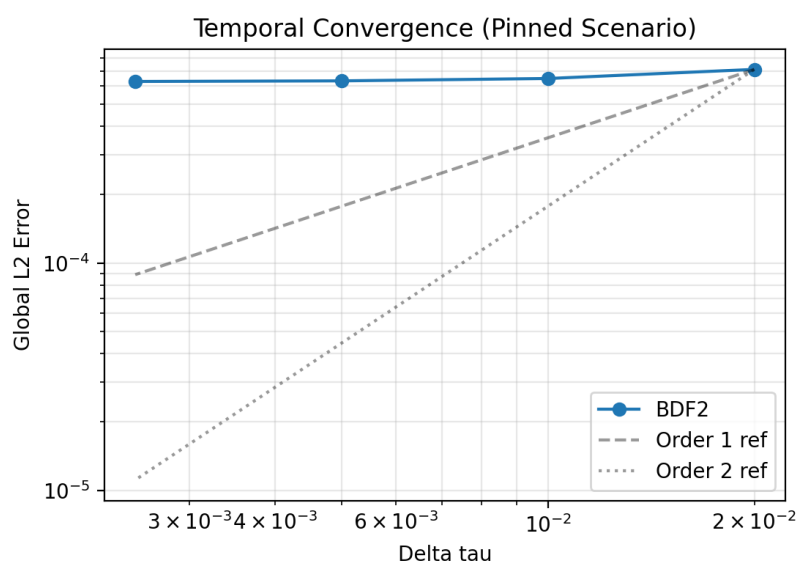

Figure 1: Temporal convergence (pinned synthetic case). Spatial error limits slope at small $\Delta\tau$.

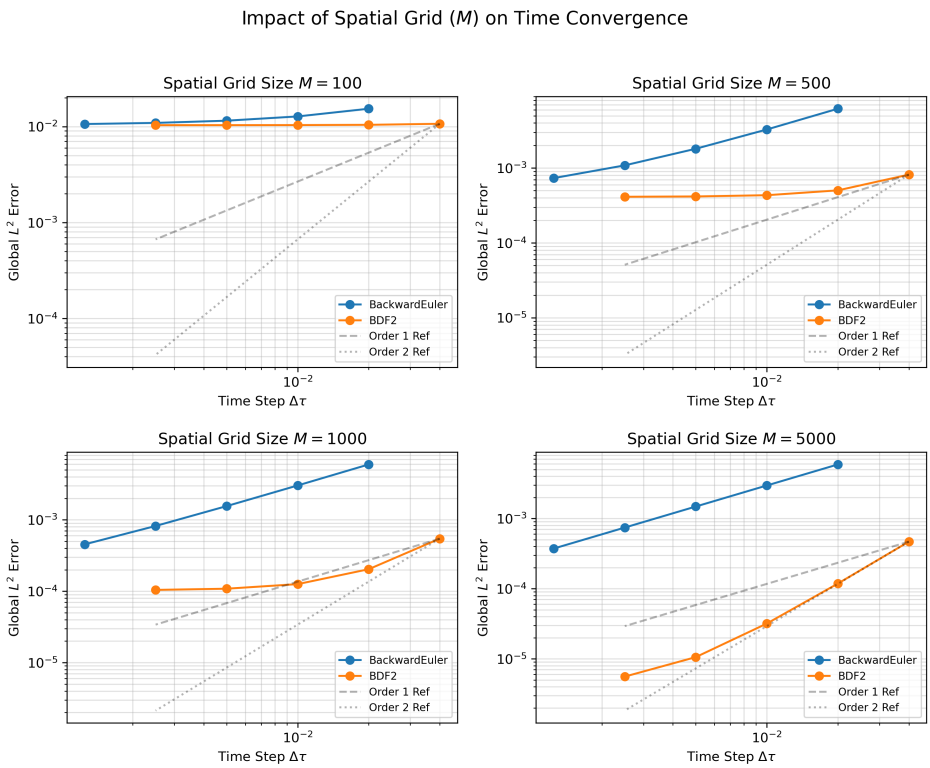

Figure 2: Effect of spatial grid M on temporal error.

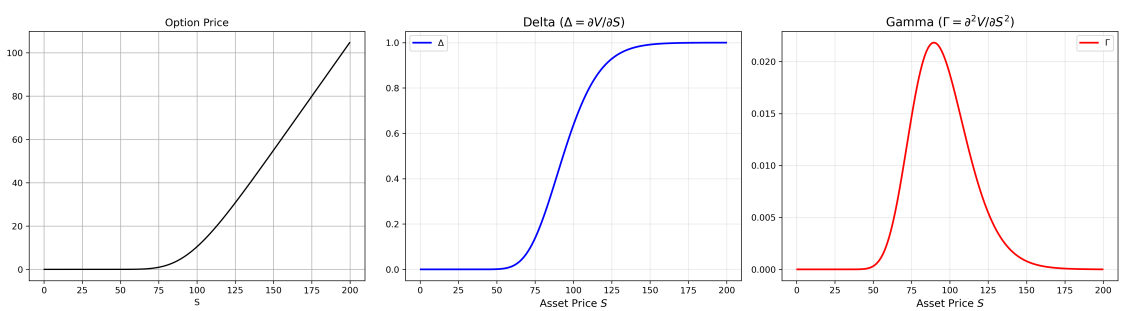

Figure 3: Price, Delta, and Gamma computed with BDF2 at $M = 1000$, $N = 500$.

References

[1] E. Süli and D. F. Mayers, *An Introduction to Numerical Analysis*, Cambridge University Press, 2003.

[2] F. Black and M. Scholes, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, 81(3):637–654, 1973.